

HDOC 25

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1. Algorithm

1. Start main program.
2. Display the menu with options 1 to 5 to the user.
3. Read the user's choice.
4. Use a switch statement to call the respective function or exit if choice is 5.
5. Function **decreasingStar()**:
 - Read n. Validate $n > 0$. Loop i from n down to 1. Inside, loop j from 1 to i. Print * . After inner loop, print newline.
6. Function **decreasingNaturalNumber()**:
 - Read n. Validate $n > 0$. Loop i from n down to 1. Inside, loop j from 1 to i. Print j . After inner loop, print newline.
7. Function **decreasingRepeatedNumber()**:
 - Read n. Validate $n > 0$. Loop i from n down to 1. Inside, loop j from 1 to i. Print i . After inner loop, print newline.
8. Function **reverseNumberTriangle()**:
 - Read n. Validate $n > 0$. Loop i from n down to 1. Inside, loop j from i down to 1. Print j . After inner loop, print newline.
9. Repeat steps 2-4 until choice == 5.
10. Stop

2. Test-Case Table

TC ID	Scenario Description	Inputs	Expected Output
TC-01	Right-angled triangle (Valid)	choice: 1, n: 4	Row 1: * * * *, Row 2: * * *, Row 3: * *, Row 4: *
TC-02	Decreasing natural-number (Valid)	choice: 2, n: 4	Row 1: 1 2 3 4, Row 2: 1 2 3, Row 3: 1 2, Row 4: 1
TC-03	Decreasing repeated-number (Valid)	choice: 3, n: 4	Row 1: 4 4 4 4, Row 2: 3 3 3, Row 3: 2 2, Row 4: 1
TC-04	Reverse-number triangle (Valid)	choice: 4, n: 4	Row 1: 4 3 2 1, Row 2: 3 2 1, Row 3: 2 1, Row 4: 1
TC-05	Invalid dimension (Boundary)	choice: 2, n: -1	Error message: Positive integer required.
TC-06	Invalid menu choice	choice: 7	Invalid choice. Please select 1-5.

3. C Program Source Code

```
#include <stdio.h>

// Function Prototypes void
decreasingStar(void);
void decreasingNaturalNumber(void); void
decreasingRepeatedNumber(void); void
reverseNumberTriangle(void);

int main() {
    int choice; do {
        printf("\n--- Decreasing Patterns Menu ---\n"); printf("1. Decreasing right-angled
        triangle\n"); printf("2. Decreasing natural-number pattern\n"); printf("3. Decreasing
        repeated-number pattern\n"); printf("4. Reverse-number triangle\n");
        printf("5. Exit\n"); printf("Enter your choice:
        "); scanf("%d", &choice);

        switch (choice) {
            case 1: decreasingStar(); break;
            case 2: decreasingNaturalNumber(); break; case 3:
            decreasingRepeatedNumber(); break; case 4:
            reverseNumberTriangle(); break;
            case 5: printf("Exiting program...\n"); break;
            default: printf("Error: Invalid choice. Please enter 1-5.\n");
        }
    } while (choice != 5);

    return 0;
}

void decreasingStar(void) { int n, i, j;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; } for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++) { printf("* ");
        }
        printf("\n");
    }
}

void decreasingNaturalNumber(void) { int n, i, j;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; } for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++) { printf("%d ", j);
```

```
        }
        printf("\n");
    }
}

void decreasingRepeatedNumber(void) {
    int n, i, j;
    printf("Enter number of rows n:");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; }
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++) {
            printf("%d ", i);
        }
        printf("\n");
    }
}

void reverseNumberTriangle(void) {
    int n, i, j;
    printf("Enter number of rows n:");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; }
    for (i = n; i >= 1; i--) {
        for (j = i; j >= 1; j--) {
            printf("%d ", j);
        }
        printf("\n");
    }
}
```